Social Media and Student Well-Being: Second Interim Report

Abstract

This report examines the relationship between social media usage and student well-being in a cross-sectional sample of 705 university-aged respondents (16–25 years). Using a cleaned Kaggle dataset, we test eight hypotheses spanning academic performance, sleep, addiction, platform type, relationship status, and gender. Exploratory data analysis reveals strong bivariate associations, formalised through OLS regression, Logit and Probit modelling, t-tests, one-way ANOVA, mediation analysis, and a Davidson–MacKinnon J-test. Six of eight hypotheses are supported or partially supported. Daily social media usage is a strong predictor of both self-reported academic impact and reduced sleep duration; sleep partially mediates the usage-to-grades pathway; and short-form video platforms (TikTok, Instagram) are associated with significantly higher addiction scores. The Logit classifier achieves 91 % accuracy (AUC = 0.96) in predicting self-reported academic impact. Hypotheses on relationship status and gender are not supported; one hypothesis on time-of-day usage could not be tested due to data limitations.

1 Topic and Motivation

Social media has become an indispensable part of daily life for students aged 16–25. Platforms such as TikTok, Instagram, and Snapchat are engineered for sustained engagement, raising legitimate concerns about their impact on academic performance, sleep quality, and mental health. Despite a growing body of literature, most existing studies focus on narrow age ranges, single-country contexts, or self-report measures without accompanying quantitative outcomes.

This project is motivated by lived experience: as students, we observe the effects of heavy social media use daily—late-night scrolling, shortened attention spans, and the diffuse anxiety of constant connectivity. We aim to translate these observations into rigorous, data-driven analysis. Specifically, we examine:

- how much time students spend on social media and which platforms dominate;
- how usage patterns correlate with academic performance and sleep duration;
- whether platform type (short-form video vs. other) modulates addiction risk;
- whether demographic factors (gender, relationship status) differentiate usage or outcomes.

Findings are intended to be useful to students preparing for high-stakes examinations, school psychologists, university welfare offices, and researchers working on digital-health interventions.

2 Aims, Objectives, and Research Questions

2.1 Central Aim

To determine *whether, how, and to what degree* social media usage is associated with student well-being outcomes, using a quantitative cross-sectional approach.

2.2 Research Questions

RQ1. Does spending more time on social media correlate with lower self-reported academic performance?
RQ2. Does greater daily use predict shorter sleep duration?
RQ3. Is the usage–performance relationship partly mediated by reduced sleep?
RQ4. Are short-form video platforms more strongly associated with addiction and poor sleep?
RQ5. Does relationship status relate to time spent online?
RQ6. Do students using social media primarily late at night report worse outcomes? *(Data limitation: untestable.)*
RQ7. Do female students report worse well-being effects at equivalent usage levels?
RQ8. Do students with higher addiction scores experience more interpersonal conflicts?

2.3 Hypotheses

Each hypothesis is stated directionally and corresponds to one research question.

H1. Students who spend more hours per day on social media report lower academic performance than those who spend less time.

H2. Higher daily social media usage is negatively correlated with average sleep duration.

H3. The negative effect of social media usage on academic performance is partly explained by reduced sleep (partial mediation): more screen time leads to less sleep, which in turn leads to worse grades.

H4. Short-form video platforms (TikTok, Instagram) are more strongly associated with high addiction scores and low grades than long-form or text-based platforms.

H5. Students in a relationship spend fewer hours on social media per day than single students.

H6. Students who use social media primarily late at night report significantly worse sleep and lower grades than those using it earlier in the day. [*Untestable: no time-of-day variable in the dataset.*]

H7. Female students are more likely to report negative well-being effects from social media use than male students, even after controlling for total usage hours.

H8. Students with higher addiction scores report more conflicts with family or partners related to social media use.

3 Literature Review

Our analytical framework is grounded in three peer-reviewed venues, each contributing a distinct theoretical lens.

Computers in Human Behavior (Elsevier) provides the primary backbone for our work on distraction, social comparison, and academic outcomes. Studies in this journal consistently show that passive social media consumption—scrolling feeds without direct interaction—is more strongly associated with negative affect and lower academic engagement than active use.

International Journal of Mental Health and Addiction (Springer) informs our operationalisation of problematic social media use. The Bergen Social Media Addiction Scale conceptualises addiction across six dimensions (salience, mood modification, tolerance, withdrawal, conflict, relapse), mapping closely onto the `Addicted_Score` variable in our dataset.

Cyberpsychology, Behavior, and Social Networking (Mary Ann Liebert) grounds our platform-specific hypotheses (H4) and our examination of how online behaviour displaces sleep. Research published here documents the unique properties of algorithmically curated short-form video feeds—infinite scroll, variable reward, and high stimulation density per second—believed to make platforms such as TikTok disproportionately habit-forming relative to earlier social media formats.

4 Data Description

4.1 Source and Structure

We use an anonymised cross-sectional dataset sourced from Kaggle, titled *Students Social Media Addiction*. The raw dataset contains 705 student records across 13 variables. Each row represents one respondent; there are no repeated measures.

4.2 Ethical and Geopolitical Exclusions

All records originating from Russia or Belarus were systematically excluded, and entries referencing VKontakte were likewise removed. This exclusion resulted in the removal of a small number of rows. The final cleaned dataset is used for all analyses reported below.

4.3 Variables

Table 1 describes all 13 variables, their type, and their role in the analysis.

Table 1: Dataset variables and analytical roles.

Variable	Type	Scale	Role / Notes
Student_ID	Identifier	—	Index; excluded from all models
Age	Numeric	16–30	Demographic control
Gender	Categorical	Male/Female	Dummy-coded (*Female* = 1)
Academic_Level	Categorical	High/Under/Grad	Dummy-coded
Country	Categorical	110 countries	Used for filtering; not modelled
Most_Used_Platform	Categorical	10 platforms	Grouped into *Short-Form* vs. *Other*
Avg_Daily_Usage_Hours	Numeric	0.5–9.9 hrs	Primary predictor across all models
Sleep_Hours_Per_Night	Numeric	4–9 hrs	Outcome (H2) and mediator (H3)
Mental_Health_Score	Numeric	1–10	Well-being indicator
Addicted_Score	Numeric	1–10	Addiction severity; predictor for H8
Affects_Academic_Performance	Binary	Yes/No	Primary outcome (H1, H3); coded 0/1
Conflicts_Over_Social_Media	Numeric	0–20	Outcome for H8
Relationship_Status	Categorical	Single/Partnered/Complicated	Grouping variable for H5

4.4 Data Quality

The dataset is exceptionally clean: zero missing values, zero duplicate rows, and all numeric variables fall within their stated plausible ranges. No imputation was required. The `Student_ID` column was set as the DataFrame index and removed from all models.

4.5 Descriptive Statistics

Table 2: Descriptive statistics for numeric variables ($n = 705$ after exclusions).

Variable	Mean	SD	Min	Q1	Median	Max
Age	20.7	2.1	18	19	21	24
Avg_Daily_Usage_Hours	4.6	2.3	0.5	2.8	4.9	9.9
Sleep_Hours_Per_Night	6.1	1.2	4.0	5.1	6.2	9.0
Mental_Health_Score	5.4	2.5	1	3	5	10
Addicted_Score	6.4	2.2	1	5	7	10
Conflicts_Over_Social_Media	3.1	2.8	0	1	3	20

Several features of the descriptive statistics warrant attention. Mean daily usage of 4.6 hours is substantial, with the upper tail reaching 9.9 hours. Mean sleep of 6.1 hours falls below the 7–9 hour recommendation for this age group. Addiction scores skew toward higher values (mean 6.4, median 7), indicating that most respondents self-report moderate-to-high addictive engagement. Instagram and TikTok are the two most-used platforms, and the sample spans approximately 110 unique countries with no single country dominating.

5 Exploratory Data Analysis

All visualisations were produced in Python using `pandas`, `matplotlib`, and `seaborn`.

5.1 Univariate Distributions

Age is roughly uniformly distributed across 18–24, reflecting the college-aged target population. Daily usage is spread across the full 0.5–10 hour range with a slight right tail. Sleep hours cluster around 5–7, with a substantial proportion of students sleeping below the clinically recommended threshold. Addiction scores are markedly right-skewed, with most students rating themselves 6–9 out of 10. Instagram and TikTok dominate platform usage, followed by Snapchat and Twitter.

[Figure 1: Three-panel histogram of Age, Avg_Daily_Usage_Hours, and Sleep_Hours_Per_Night.]

[Figure 2: Frequency distributions of Mental_Health_Score, Addicted_Score, and Conflicts_Over_Social_Media.]

[Figure 3: 2×2 bar chart grid for Gender, Academic Level, Most Used Platform, and Relationship Status.]

[Figure 4: Top-15 countries by respondent count (horizontal bar chart).]

5.2 Bivariate Analysis

5.2.1 H1 — Usage and Academic Performance

Students who report academic impact use social media for a median of 5.5 hours per day, compared to 3.9 hours for those who do not—a difference of 1.6 hours. Density distributions for the two groups are clearly separated, with the "affected" group shifted substantially to the right.

[Figure 5: Boxplot and overlaid density of Avg_Daily_Usage_Hours by Affects_Academic_Performance (Yes/No).]

5.2.2 H2 — Usage and Sleep

The Pearson correlation between daily usage and sleep is $r = -0.79$, one of the strongest pairwise relationships in the dataset. Each additional hour of social media use is associated with meaningfully shorter sleep, as shown by the downward-sloping OLS regression line.

[Figure 6: Scatter plot of Avg_Daily_Usage_Hours vs. Sleep_Hours_Per_Night with linear fit ($r = -0.79$).]

5.2.3 H4 — Platform Type and Addiction

TikTok and Instagram users cluster at higher addiction scores, while users of text-based and long-form platforms (Twitter, YouTube, LinkedIn) display lower medians.

[Figure 7: Side-by-side boxplots of Addicted_Score and Avg_Daily_Usage_Hours by Most_Used_Platform.]

5.2.4 H5 — Relationship Status and Usage

The three relationship-status groups are virtually indistinguishable: Single (4.80 hrs), In Relationship (4.70 hrs), Complicated (4.75 hrs). No visual support for H5 is apparent.

[Figure 8: Boxplot of Avg_Daily_Usage_Hours by Relationship_Status.]

5.2.5 H7 — Gender and Well-being

Median values across gender groups are nearly identical for daily usage, mental health score, and sleep duration. The conditional gender effect predicted by H7 requires regression interaction terms, tested in the modelling stage below.

[Figure 9: Three-panel boxplot of Usage Hours, Mental Health Score, and Sleep Hours by Gender.]

5.2.6 H8 — Addiction and Conflicts

The correlation between addiction score and conflicts is $r = 0.93$—the strongest pairwise relationship in the dataset—with a near-linear positive association plainly visible in the scatter plot.

[Figure 10: Scatter plot (with jitter) of Addicted_Score vs. Conflicts_Over_Social_Media with OLS regression line.]

5.3 Correlation Structure

Table 3 summarises the key pairwise correlations; the full annotated heatmap is presented in Figure 11.

Table 3: Key Pearson correlations between numeric variables.

Variable A	Variable B	r	Relevance
Avg_Daily_Usage_Hours	Sleep_Hours_Per_Night	−0.79	H2
Addicted_Score	Conflicts_Over_Social_Media	+0.93	H8
Addicted_Score	Mental_Health_Score	−0.95	General well-being
Avg_Daily_Usage_Hours	Addicted_Score	+0.60	General thesis
Avg_Daily_Usage_Hours	Mental_Health_Score	−0.55	General thesis
Age	Avg_Daily_Usage_Hours	−0.05	Negligible

[Figure 11: 6×6 annotated Pearson correlation heatmap (RdBu_r palette, centred at zero).]

6 Methods

Statistical analysis was conducted in Python using `statsmodels 0.14` and `scikit-learn 1.4`. All tests use a significance threshold of $\alpha = 0.05$ (two-tailed) unless stated otherwise.

6.1 t-tests and ANOVA

Welch's two-sample t-test was used to compare group means for H1, H4b, and H7. Welch's variant does not assume equal variances, making it more robust under heteroscedasticity and unequal sample sizes.

One-way ANOVA was applied for H4a (addiction score across all platforms) and H5 (usage hours across the three relationship-status groups). ANOVA tests the null hypothesis that all group means are equal.

6.2 OLS Regression

Two ordinary least-squares models were estimated:

1. **Model S (H2 / H3 Step A).** Sleep duration as the dependent variable; predictors: usage hours, age, gender dummy, and academic-level dummies.
2. **Model C (H8).** Conflict count as the dependent variable; predictors: addiction score, age, and gender dummy.

Both models take the standard form

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

6.3 Logit and Probit Models

Because the primary academic outcome (`Affects_Academic_Performance`) is binary, we model the probability of a positive response as

$$P(y_i = 1 \mid \mathbf{x}_i) = F\big(\mathbf{x}_i^\top \boldsymbol{\beta}\big)\,, \tag{2}$$

where $F(\cdot)$ is the logistic CDF (Logit) or the standard normal CDF (Probit). Two Logit specifications are estimated:

- **Logit A (H1).** Predictors: usage hours, age, gender, academic level, and a short-form platform dummy.

- **Logit B (H3).** Logit A augmented with `Sleep_Hours_Per_Night` to test mediation.

Marginal effects, evaluated at the mean, are reported alongside coefficients for substantive interpretation.

6.4 Mediation Analysis (H3)

We follow the Baron–Kenny sequential regression approach:

1. **Step A.** Regress the mediator (sleep) on the treatment (usage) — Model S.
2. **Step B.** Regress the outcome (academic impact) on the treatment without the mediator — Logit A.
3. **Step C.** Regress the outcome on both treatment and mediator — Logit B.

Partial mediation is inferred if the usage coefficient in Step C is smaller in magnitude than in Step B; full mediation if it becomes non-significant.

6.5 Model Comparison: J-test

The Davidson–MacKinnon J-test is applied to compare the non-nested Logit and Probit specifications. In J-test 1, fitted probabilities from the Probit are added to the Logit specification; in J-test 2, the reverse. Significance of the added predictions indicates that the alternative model captures information not present in the primary model.

6.6 Predictive Evaluation

The Logit model is evaluated on a held-out test set using an 80/20 stratified train–test split. Evaluation metrics include a confusion matrix (at threshold $\tau = 0.5$), precision, recall, F_1-score per class, and the ROC curve with its area under the curve (AUC).

7 Results

7.1 H1 — Usage and Academic Performance

A Welch t-test comparing daily usage between students who report academic impact ($n \approx 352$, $\bar{x} = 5.9$ hrs, $s = 1.9$) and those who do not ($n \approx 353$, $\bar{x} = 3.3$ hrs, $s = 1.7$) yields $t = 21.5$, $p \approx 3 \times 10^{-99}$. The difference is highly significant. Logit A confirms this at the multivariate level: the marginal effect of one additional hour of daily usage on the probability of reporting academic impact is statistically significant after controlling for age, gender, academic level, and platform type. **H1 is supported.**

7.2 H2 — Usage and Sleep

OLS Model S explains 66 % of the variance in sleep duration ($R^2 = 0.66$, $F = 271.2$, $p \approx 5 \times 10^{-161}$). Each additional hour of daily social media use is associated with a reduction of approximately 0.38 hours in sleep, controlling for demographics. **H2 is strongly supported.**

7.3 H3 — Sleep as Mediator of Usage on Academic Performance

Following the Baron–Kenny framework:

- **Step A.** Established in H2: usage significantly predicts sleep.
- **Step B (Logit A).** Usage coefficient $\hat{\beta} = 2.99$.
- **Step C (Logit B).** Adding sleep reduces the usage coefficient to $\hat{\beta} = 1.88$, a reduction of 37 %.

The 37 % attenuation indicates *partial* mediation: sleep explains part of the mechanism through which heavy usage damages grades, but a substantial direct effect persists. **H3 is partially supported.**

7.4 H4 — Short-Form Platforms and Addiction

One-way ANOVA across all platforms yields $F = 35.7$, $p \approx 1.5 \times 10^{-60}$, confirming that addiction scores differ significantly across platforms. A follow-up Welch t-test contrasting short-form (TikTok, Instagram) against all other platforms yields $t = 9.1$, $p \approx 1.3 \times 10^{-18}$, with short-form users reporting meaningfully higher addiction scores. **H4 is supported.**

7.5 H5 — Relationship Status and Usage

One-way ANOVA yields $F = 0.41$, $p = 0.66$. Group means are indistinguishable: Single (4.80 hrs), In Relationship (4.70 hrs), Complicated (4.75 hrs). **H5 is not supported.**

7.6 H6 — Late-Night Usage

The dataset contains no time-of-day variable. H6 cannot be evaluated with the available data. Future research designs should record usage by time window (e.g., before vs. after midnight). **H6 is untestable.**

7.7 H7 — Gender Differences in Well-being

Welch t-tests for `Mental_Health_Score`, `Avg_Daily_Usage_Hours`, and `Sleep_Hours_Per_Night` between male and female respondents all yield $p > 0.05$; median values differ by less than 0.1 across all three outcomes. Regression models

including a gender × usage interaction term likewise fail to reach significance. **H7 is not supported.**

7.8 H8 — Addiction and Conflicts

OLS Model C achieves $R^2 = 0.874$ ($F \gg 1000$, $p \approx 0$). Each one-unit increase in addiction score is associated with approximately 0.55 additional conflicts, controlling for age and gender. This is the strongest explanatory model in the study. **H8 is strongly supported.**

7.9 Logit Classifier Performance

On the stratified 80/20 split, the Logit model achieves accuracy of 91 % and AUC of 0.959, with balanced precision and recall across both classes. These results confirm that daily usage hours, combined with basic demographic controls, are strong and generalisable predictors of self-reported academic impact.

[Figure 12: Confusion matrix and ROC curve for the held-out test set.]

7.10 Logit vs. Probit: J-test

Both cross-tests are marginally significant, indicating that neither model strictly dominates. The Probit attains a slightly lower AIC (386.0 vs. 390.1) and marginally higher pseudo-R^2 (0.595 vs. 0.591). In substantive terms the two models are interchangeable; the Logit is retained as the primary model for the interpretability of its odds ratios.

8 Conclusions

8.1 Hypothesis Summary

Table 4: Hypotheses, methods, and verdicts.

H	Claim	Method(s)	Verdict
H1	More usage → lower academic performance	t-test; Logit A	**Supported**
H2	More usage → less sleep	OLS ($R^2 = 0.66$)	**Supported**
H3	Sleep partially mediates usage → grades	Logit mediation (−37%)	**Partially Supported**
H4	Short-form platforms more addictive	ANOVA; t-test	**Supported**
H5	Partnered students use SM less	ANOVA ($p = 0.66$)	**Not Supported**
H6	Late-night usage → worse outcomes	—	*Untestable*
H7	Females report worse effects	t-tests ($p > 0.05$)	**Not Supported**
H8	Higher addiction → more conflicts	OLS ($R^2 = 0.87$)	**Supported**

8.2 Key Findings

1. **Daily usage is the dominant predictor.** Usage hours is the single most important variable across every model estimated. Its associations with academic impact, sleep deprivation, and addiction are statistically robust and practically meaningful.
2. **Sleep is a partial mediator.** Approximately 37 % of the effect of social media use on academic performance operates through sleep displacement. Interventions targeting late-night device use could therefore address both sleep quality and academic outcomes simultaneously.
3. **Addiction and conflicts are nearly linearly related.** $R^2 = 0.87$ for the conflicts model is striking: addiction score captures almost all of the explainable variance in interpersonal conflict, suggesting that reducing addictive usage patterns may yield significant downstream benefits for social relationships.
4. **Short-form platforms carry elevated addiction risk.** TikTok and Instagram users report addiction scores that are statistically and practically higher than users of other platforms. The algorithmic design of these platforms—infinite scroll, variable-ratio reinforcement—may be the key mechanism.
5. **Gender and relationship status do not differentiate outcomes.** Neither variable, univariately or in interaction with usage, produces significant effects. The risks associated with heavy social media use appear to be broadly uniform across gender and relationship status.
6. **The Logit classifier generalises well.** AUC of 0.96 and accuracy of 91 % on a held-out test set confirm that the model meaningfully discriminates students who self-report academic harm.

8.3 Limitations and Future Directions

Cross-sectional design. Causality cannot be inferred. Experimental or longitudinal designs are required to establish whether usage causes harm or whether students with poorer outcomes gravitate toward heavier use.

Self-report measures. Both addiction scores and academic impact are subjective. Objective GPA data or device screen-time logs would strengthen conclusions.

Absent variables. The dataset lacks time-of-day usage data (preventing H6 testing), objective sleep measures

(actigraphy), and clinical mental health diagnoses.

Geographic heterogeneity. With 110 countries and small per-country samples, cultural context is inadequately controlled. Country-level fixed effects would require a larger and more geographically balanced sample.

Platform operationalisation. The binary Short-Form/Other split is a simplification. Future analyses should apply a finer-grained taxonomy and account for multi-platform users.

Appendix: Figure Placement Guide

Table 5 maps each figure placeholder to its source code block and the file name to be inserted. To insert a figure, replace the corresponding *[Figure X: ...]* placeholder in the LaTeX source with a standard `\includegraphics` command, for example:

```
\includegraphics[width=\linewidth]{figures/fig_h2_sleep.png}
```

Table 5: Visualisation placement reference.

Fig.	Section	File to insert	Source
1	5.1	`fig_univar_continuous.png`	Stage 2, Sec. 4, histogram block 1
2	5.1	`fig_univar_scores.png`	Stage 2, Sec. 4, histogram block 2
3	5.1	`fig_categorical.png`	Stage 2, Sec. 4, 2×2 subplot
4	5.1	`fig_countries.png`	Stage 2, Sec. 4, barh countries
5	5.2.1	`fig_h1_academic.png`	Stage 2, Sec. 5, H1 block
6	5.2.2	`fig_h2_sleep.png`	Stage 2, Sec. 5, H2 scatter
7	5.2.3	`fig_h4_platforms.png`	Stage 2, Sec. 5, H4 boxplots
8	5.2.4	`fig_h5_relationship.png`	Stage 2, Sec. 5, H5 boxplot
9	5.2.5	`fig_h7_gender.png`	Stage 2, Sec. 5, H7 boxplots
10	5.2.6	`fig_h8_conflicts.png`	Stage 2, Sec. 5, H8 scatter
11	5.3	`fig_heatmap.png`	Stage 2, Sec. 6, heatmap
12	7.8	`fig_cm_roc.png`	Stage 3, Sec. 6, CM + ROC block